Vishesh Konduru

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Education

Worcester Polytechnic Institute — Worcester, MA

Master of Science in Robotics Engineering — *Dec 2027*

- **Coursework**: Robot Dynamics, Robot Control, Swarm Intelligence, Biomedical Robotics, Imaging for Medical Robotics

Bachelor of Science in Robotics Engineering, Minor in Computer Science | GPA: 3.81 — *Dec 2027*

- **Coursework**: Unified Robotics I–IV, Control Engineering, Embedded Systems, Algorithms, Software Engineering

Experience

Cavall Labs (CMU VentureBridge) — Mar 2026 – Present

Software Engineer — *Remote*

- Developed Heimdall, a multi-agent system that extracts molecules from images and patents into SMILES through vision LLM sub-agents that transcribe drawings and a deterministic RDKit backend that assembles and validates every structure
- Designed an agent role system where a centralized config specifies each role's model, effort, and tool permissions and is compiled into native agent definition files for 5 AI coding agents, Claude Code, Codex, Cursor, Zed, and OpenCode
- Building a browser-based molecule sketcher in TypeScript and RDKit WASM that turns raw pen strokes into bonds, rings, stereo wedges, and handwritten element labels, with a web worker recognizer and live SMILES output
- Co-first authored *Heimdall: An Agentic Eye for Chemistry*, accepted to the NeurIPS 2026 ML4Molecules workshop

Worcester Polytechnic Institute — Feb 2024 – May 2026

Robotics Outreach Assistant — *Worcester, MA*

- Built a pandas pipeline over multi-year applicant and alumni records to model trends that shaped Admissions outreach
- Organized outreach events and produced Admissions social media content, raising participation 70% and engagement 90%

Projects

Autonomous Robotic Design for Strategic Adversarial Games | *Major Qualifying Project*

- Designed and built a fully autonomous robot on a ROS 2 stack to play the VEX Push Back game, using Nav2 MPPI for path planning, AMCL fusing dual LiDARs, IMU, and odometry for localization, and BehaviorTree.CPP for match strategy
- Built a four camera perception pipeline on a Jetson Nano with Retinex based illumination correction in OpenCV, streaming a stitched 2560x400 feed over TCP to a Jetson AGX Orin running YOLOv11 detection at 30 Hz
- Trained a multi-camera occupancy model with a shared ResNet18 backbone and a Transformer encoder that cross-attends 48x48 learnable bird's eye view queries to image features from all four cameras, using 200,000 synthetic samples
- Developed a PPO self-play strategy agent in a custom continuous-space game simulation with shaped rewards for scoring and zone control, trained over 26M games on 102 CPU cores to a converged TrueSkill rating of about 160
- Designed the robot's mechanical systems around tight packaging constraints, fitting an X-drive holonomic drivetrain, a roller intake scoring at all three goal heights, and a 10 piece magazine hopper alongside two Jetsons, dual LiDARs, and five cameras

Autonomous Mapping and Localizing Robot

- Implemented frontier-based exploration on a TurtleBot3 to autonomously map an unknown maze as an occupancy grid using LiDAR SLAM, evaluating both gmapping and Cartographer as mapping backends
- Programmed A* path planning with non-blocking navigation, AMCL self-localization, and centroid-based frontier clustering

Custom Variable-Pitch Ultrasound Transducer Beamforming for Needle Guidance

- Developed a MATLAB beamformer for the JAR526, a custom 128-element variable-pitch ultrasound transducer probe with a center slit for ultrasound-guided needle insertion, implementing DAS, DMAS, SLSC, and coherence factor (CF) reconstruction
- Programmed a Verasonics research ultrasound system with per-element geometry for live plane-wave imaging, validating image quality in Field II simulation and on tissue phantoms with roughly 2x sharper lateral resolution

Giving Shells a New Life: Oyster Waste Solutions in Angsila, Thailand | *Interactive Qualifying Project*

- Collaborated on a 7 person interdisciplinary team with Chulalongkorn University, interviewing 24 shuckers and officials
- Nominated for WPI's President's IQP Award and the Forum on Education Abroad's Award for Academic Achievement Abroad
- Designed a scalable production pipeline staged along NASA Technology Readiness Levels from low-tech shucking house processing to an industrial plant, turning 35 tonnes per week of shells into rope and concrete free raw material
- Fabricated paver, tile, and coaster prototypes in hydraulic press molds and ran flexural, water absorption, porosity, and density tests on 0 to 75% shell-concrete composites, measuring strength from 2.8 to 10.3 MPa to match formulations to applications

Technical Skills

Languages: Python, Java, C/C++, MATLAB, TypeScript/JavaScript, SQL
Robotics & Vision: ROS/ROS 2, Nav2, BehaviorTree.CPP, Cartographer, Gazebo, Isaac Sim, RViz, OpenCV, YOLO, PyTorch
Hardware & Tools: NVIDIA Jetson, ESP32, OpenPLC, SolidWorks, Onshape, KiCad, Git, Docker, Jupyter Notebook

Leadership

VEXU Robotics | *Secretary, Marketing Chair, Build Lead, Driver, Programmer* — Apr 2023 – Present

- Lead the construction and programming of robots competing at the VEXU World Championship, organized tasks through agile and scrum to raise team participation 70%, and volunteered 400+ hours at VEX, FRC, FTC, and FLL competitions